

Database Design (Unit II) – Detailed Notes

1. Database Design

Database design is the process of organizing data in a database efficiently so that redundancy is minimized and integrity is maintained.

Objectives:

- Reduce data redundancy
- Maintain data consistency
- Ensure data integrity
- Improve data access speed
- Make database easy to maintain

Steps:

1. Requirement Analysis
2. Conceptual Design (ER Model)
3. Logical Design (Relational Model)
4. Physical Design

2. Integrity Constraints

Integrity constraints are rules that maintain the accuracy and consistency of data in a database.

Types:

- Domain Constraint
- Referential Integrity

2.1 Domain Constraints

A domain constraint specifies that each attribute must contain values from a specific domain (set of values).

Example:

Student Table

RollNo | Name | Age

101 | Aman | 21

102 | Ravi | 20

Age must be numeric. A value like 'ABC' would violate the domain constraint.

2.2 Referential Integrity

Referential integrity ensures that a foreign key value must exist in the primary key of another table.

Example:

Student Table

StudentID | Name

1 | Aman

2 | Ravi

Course Table

StudentID | Course

1 | MCA

2 | BCA

If StudentID = 5 appears in Course table but not in Student table, integrity is violated.

3. Functional Dependency

Functional dependency describes the relationship between attributes.

Notation: $A \rightarrow B$

Meaning: Attribute A determines attribute B.

Example:

RollNo $\rightarrow$ Name

RollNo $\rightarrow$ Course

Types of Functional Dependency

Partial Dependency:

Occurs when a non-key attribute depends on part of a composite key.

Transitive Dependency:

Occurs when a non-key attribute depends on another non-key attribute.

4. Normalization

Normalization is the process of organizing tables to reduce redundancy and dependency problems.

Advantages:

- Reduces duplication
- Improves consistency
- Saves storage
- Simplifies database structure

5. Second Normal Form (2NF)

A relation is in 2NF if:

1. It is already in 1NF
2. No partial dependency exists

Example:

StudentID | CourseID | StudentName

Solution:

Student(StudentID, StudentName)

Course(StudentID, CourseID)

6. Third Normal Form (3NF)

A table is in 3NF if:

1. It is in 2NF
2. No transitive dependency exists

Example:

StudentID | DeptID | DeptName

Solution:

Student(StudentID, DeptID)

Department(DeptID, DeptName)

7. Boyce-Codd Normal Form (BCNF)

BCNF is a stronger version of 3NF.

For every dependency $A \rightarrow B$, A must be a super key.

8. Multivalued Dependency

Occurs when one attribute determines multiple independent values.

Notation: $A \twoheadrightarrow B$

Example:

Student | Hobby | Language

Aman | Cricket | English

Aman | Music | Hindi

9. Fourth Normal Form (4NF)

A table is in 4NF if:

- It is in BCNF
- It has no multivalued dependency

Solution:

StudentHobby(Student, Hobby)

StudentLanguage(Student, Language)

10. Join Dependency and Fifth Normal Form (5NF)

Join dependency occurs when a table can be reconstructed by joining multiple smaller tables.

A relation is in 5NF if:

- It is already in 4NF
- No join dependency exists.